

Rebuild reliability: Voronoi vs PRM vs Visibility Graph

Voronoi's rebuilds almost always find a route: 3 % of attempts fail at event @ 20 ms, against 56 % for PRM and 34 % for Visibility Graph. Total over all 1,400 moving-obstacle runs per planner (policy comparison and horizon sweep): planning time and failed attempts, Voronoi 343 s, 1,333 failed; PRM 447 s, 17,360 failed; Visibility Graph 857 s, 19,263 failed. A failed attempt leaves the robot on its previous route; an escape move steps a robot out of an obstacle zone and is not a rebuild. In each group, ▲ blue (bold) marks the best planner and ▼ orange the worst.

	Rebuild attempts per run	Attempts that found no route (%)	Escape moves per run	Planning time per run (ms)	Runs failed: no route for 1 s
Event (ours) @ 20 ms					
Voronoi	9.4 ▲	3 ▲	7.0	123 ▲	0 ▲
PRM	17.4	56 ▼	5.1 ▲	146	0 ▲
Visibility Graph	35.0 ▼	34	8.5 ▼	469 ▼	1 ▼

Event (ours) @ 20 ms, 50 ms horizon

Voronoi	14.5 ▲	7 ▲	14.9	188 ▲	0 ▲
PRM	25.7	65 ▼	13.5 ▲	206	2
Visibility Graph	42.6 ▼	46	19.9 ▼	529 ▼	4 ▼

Cycle @ 20 ms

Voronoi	113.2	2 ▲	9.9	757 ▲	0 ▲
PRM	93.0 ▲	18 ▼	6.1 ▲	1113	3 ▼
Visibility Graph	125.2 ▼	14	14.9 ▼	1707 ▼	1